

Project Part 2

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Repository URL: <https://github.com/maryamimran12/stat3080-project>

Introduction

This report continues the analysis from Part 1, which explored how annual unemployment rates vary across U.S. states and Census regions. The exploratory analysis revealed that in 2024, the West had the highest mean unemployment rate (4.02%) and the Midwest the lowest (3.46%), with notable variation in spread across regions. This report formally tests whether those regional differences are statistically significant.

Research question: Do mean annual unemployment rates differ significantly across U.S. Census regions in 2024?

Methods

A one-way analysis of variance (ANOVA) is used to test whether at least one Census region has a different mean unemployment rate from the others in 2024. ANOVA is appropriate here because the response variable (state unemployment rate) is continuous, there are four independent groups (Northeast, South, Midwest, West), and the goal is to compare means across more than two groups using a single standalone procedure.

The hypotheses are:

$$H_0 : \mu_{NE} = \mu_S = \mu_{MW} = \mu_W$$

H_a : At least one regional mean unemployment rate differs from the others

A significance level of $\alpha = 0.05$ is used.

Assumption verification. One-way ANOVA requires (1) independence of observations, (2) approximate normality within each group, and (3) equal variances across groups.

Independence: Each observation is a U.S. state or D.C. States are distinct geographic and economic units; their unemployment rates are collected independently by the BLS LAUS program. This assumption is reasonably met.

Normality: With group sizes ranging from 9 (Northeast) to 17 (South), the Central Limit Theorem cannot be invoked. Normal Q-Q plots for each region are examined instead (Figure 1). All four plots show points tracking closely along the reference line with no severe departures, supporting approximate normality within each region.

Equal variances: The standard deviations from Part 1 (Midwest: 0.96, Northeast: 0.79, West: 0.84, South: 0.68) are all within a factor of 1.5 of one another. The residual plot (Figure 2) shows no systematic fanning or clustering across groups. The equal variance assumption is reasonably satisfied.

Figure 1: Normal Q-Q Plots by Census Region (2024)

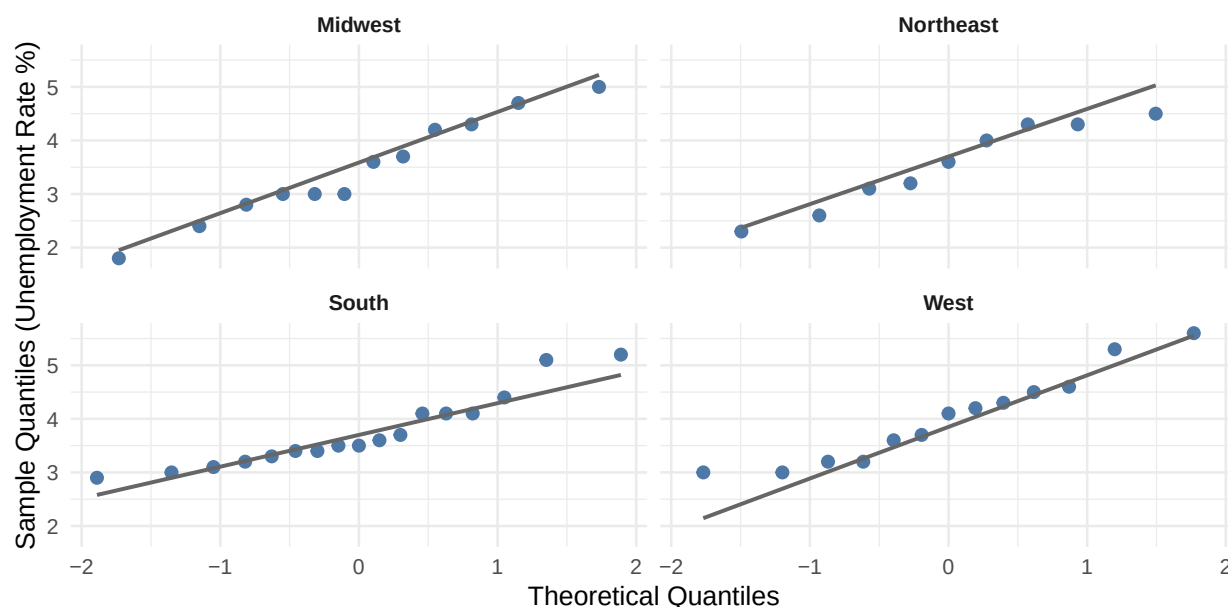


Figure 1: Normal Q-Q plots of 2024 state unemployment rates by Census region. Points track closely along the reference line in all four regions, supporting approximate normality.



Figure 2: Residuals from the one-way ANOVA model plotted against fitted values (regional means). The spread of residuals is similar across all four regions, supporting the equal variance assumption.

Results

```
# One-way ANOVA: test whether mean unemployment rate differs across Census regions
# Response: 2024 state unemployment rate (continuous)
# Factor: Census region (4 levels: Midwest, Northeast, South, West)
anova_result <- aov(Rate ~ Region, data = unemp_2024)

# Display the ANOVA table
summary(anova_result)
```

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## Region      3  2.297  0.7658   1.159  0.335
## Residuals  47 31.056  0.6608
```

The ANOVA table shows an F-statistic of 1.16 on 3 and 47 degrees of freedom, with a p-value of 0.3354. Since this p-value exceeds $\alpha = 0.05$, the result is not statistically significant. We fail to reject H_0 ; the data do not provide sufficient evidence that mean unemployment rates differ across Census regions in 2024.

Conclusions

The one-way ANOVA provides insufficient evidence that mean state unemployment rates differ across U.S. Census regions in 2024 ($F = 1.16$, $p = 0.3354$). In practical terms, while the exploratory analysis in Part 1 revealed descriptive differences in regional means — the West averaging 4.02% and the Midwest 3.46% — the within-region variability is large enough that these differences are not statistically distinguishable at the 0.05 level. Census region alone does not appear to be a statistically significant predictor of state unemployment rates in 2024.

Limitations. First, the ANOVA identifies that at least one regional mean differs but does not specify which pairs differ; a post-hoc procedure (e.g., Tukey HSD) would be needed for pairwise comparisons. Second, the analysis treats D.C. as part of the South, which may inflate that region’s mean given D.C.’s uniquely concentrated federal labor market. Third, unemployment rates are model-based estimates subject to periodic BLS revision, introducing a small degree of measurement uncertainty.

Generalizability. Because the BLS LAUS data cover all 50 states and D.C. — not a sample — the results describe the actual 2024 population of U.S. state-level labor markets. Caution is warranted in generalizing across years, as regional patterns shift with economic cycles, policy changes, and external shocks such as the COVID-19 pandemic.

References

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AI Attribution: Portions of this report including the R Markdown structure, code, and written prose were drafted with assistance from Claude (Anthropic, claude.ai). All data values were verified against official BLS publications. All output was reviewed and edited by the author in accordance with UVA academic integrity guidelines.